Practical : 1	**Set up Python environment for AI and implement a simple reflex agent in a toy environment.**
Date : 05/08/26	

Aim: -

To set up a Python environment for Artificial Intelligence and implement a simple goal-based agent in a toy environment. The agent navigates through a grid environment, cleans all dirty locations, and finally reaches the goal location.

Algorithm:-

1. Start the program.
2. Import the required libraries: random, time, os, and datetime.
3. Display the program title.
4. Display the current date, time, and operating system.
5. Randomly generate the room status as **Dirty** or **Clean**.
6. Check the room status.
7. If the room is **Dirty**, start cleaning the room.
8. Otherwise, display that no cleaning is required.
9. Display the task completion message.
10. Stop the program.

Program:-

```
# ================================================================
# Practical No. 1
# Simple Reflex Agent
# ================================================================

import random
import time
import os
import datetime

print("=" * 60)
print(" SIMPLE REFLEX AGENT")
print("=" * 60)

current_time = datetime.datetime.now()

print("\nCurrent Date :", current_time.strftime("%d-%m-%Y"))
print("Current Time :", current_time.strftime("%H:%M:%S"))

print("\nOperating System :", os.name)

room_status = random.choice(["Dirty", "Clean"])

print("\nChecking Room Status...")

time.sleep(2)

print("\nRoom Status :", room_status)
if room_status == "Dirty":
    print("\nAI Decision : Start Cleaning")
    time.sleep(2)
    print("Action : Room Cleaned Successfully")
else:
    print("\nAI Decision : No Cleaning Required")
    print("Action : Room is Already Clean")

print("\nTask Completed Successfully.")
print("=" * 60)
```

Output:-

```
C:\Users\Vishnu>cd C:\Users\Vishnu\OneDrive\AI_Practicals

C:\Users\Vishnu\OneDrive\AI_Practicals>python practical1_reflex.py
============================================================
 SIMPLE REFLEX AGENT
============================================================

Current Date : 23-08-2026
Current Time : 21:00:17

Operating System : nt

Checking Room Status...

Room Status : Dirty

AI Decision : Start Cleaning
Action : Room Cleaned Successfully

Task Completed Successfully.
============================================================
```

```
C:\Users\Vishnu\OneDrive\AI_Practicals>python practical1_reflex.py
============================================================
 SIMPLE REFLEX AGENT
============================================================

Current Date : 23-08-2026
Current Time : 21:49:15

Operating System : nt

Checking Room Status...

Room Status : Clean

AI Decision : No Cleaning Required
Action : Room is Already Clean

Task Completed Successfully.
============================================================
```

Conclusion:- The Python environment was successfully set up and Thus, the **Simple Reflex Agent was successfully implemented using Python**. The agent observes the current room status (**Dirty or Clean**) and takes an immediate action based on predefined rules. If the room is dirty, the agent cleans it; if the room is clean, no cleaning action is required.